Q What is Transaction?

- Unit of work done
  - against the database
  - done in logical sequence

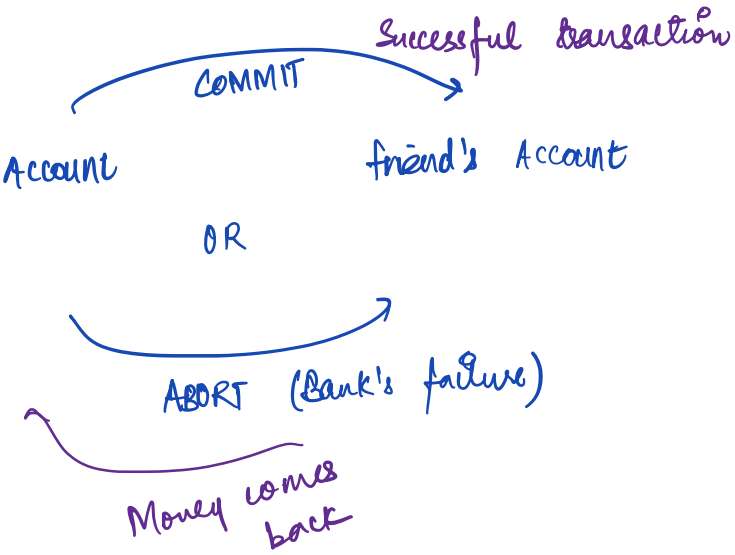
→ Can be done
- manually → SQL → Delete / update
- automated → scripted language < Python / Java

- It is propagation from one step/state to another step/state inside a database. (maintain data integrity & introduce no new database)

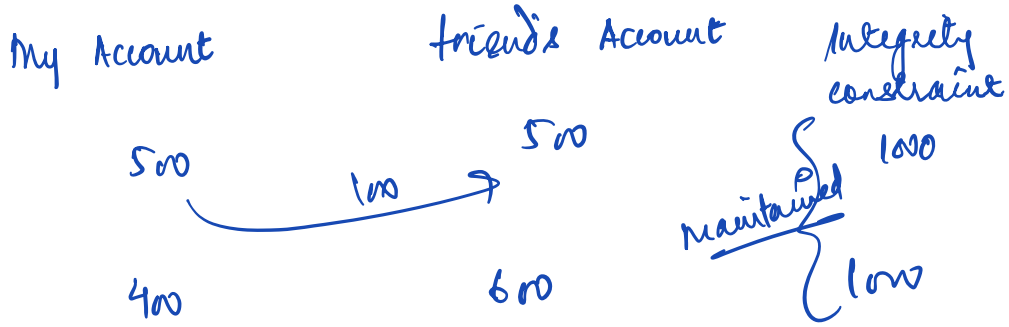
4 Properties of Transactions → ACID

- Atomicity → entire transaction takes place as at once or doesn't take place at all
- Consistency → Data in database should be consistent before & after the transaction takes place
- Isolation → If there are multiple transactions happening, they should be independent of each other.
- Durability
  ↳ A computed transaction in case of failure should persist

Atomicity



Consistency → Integrity constraints of database should be maintained before & after the transaction.



Isolation

Multiple transactions are performed independently w/o any interference

With isolation

My Account	friend's account	Parent's account
500	500	500
500	1000	0

2 transactions w/o interference.

Without isolation

My Account	friend's account	Parent's account
1000	500	500
900	100	600

100

600 → violates data consistency

If before completing transaction with friend, parent's transaction starts – instead of reading it ₹900 it still reads ₹1000

Durability

Complete and successful transaction should be recorded even in case of system failure. (COMMIT Command)

↳ To permanently store committed transaction in a database.

My Account	friend's Account
500	0
200	200

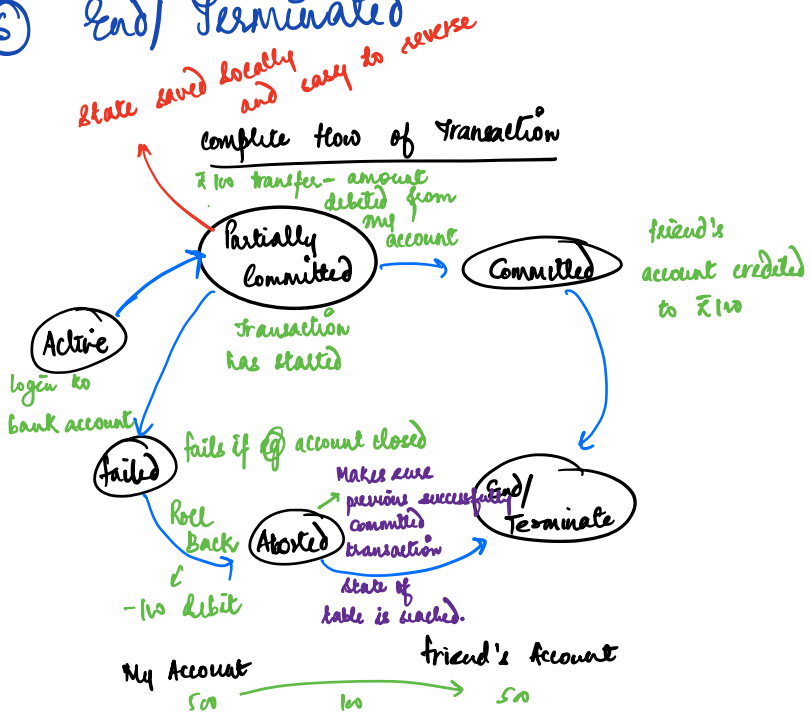
eg

If ATM → power failure

↓

still records transaction

States of Transaction

6 states

1. Active State
2. Partially committed
3. Committed
4. failed
5. Aborted
6. End/ Terminated